课程大纲
COURSE SYLLABUS

1.	**课程代码/名称** **Course Code/Title**	CSE5002/智能数据分析 Intelligent Data Analysis
2.	**课程性质** **Compulsory/Elective**	专业选修课 Major elective course
3.	**课程学分/学时** **Course Credit/Hours**	3/64（32 Lectures+32 Labs）
4.	**授课语言** **Teaching Language**	中文 Chinese
5.	**授课教师** **Instructor(s)**	袁博，助理教授，计算机科学与工程系，yuanb@sustech.edu.cn Bo Yuan, Assistant Professor, Department of Computer Science and Engineering, yuanb@sustech.edu.cn
6.	**是否面向本科生开放** **Open to undergraduates or not**	否 Not
7.	**先修要求** **Pre-requisites**	（如面向本科生开放，请注明区分内容。 If the course is open to undergraduates, please indicate the difference.） 线性代数，概率与统计 Linear Algebra, Probability and Statistics

8. 教学目标
Course Objectives

本课程介绍智能数据分析的基本思想和算法，并侧重于基本的分析任务如分类和聚类。通过一系列的理论课和实验课，本课程将涵盖多种分类和聚类算法、输出校正、模型选择、数据表征（特征分析）、排序学习、相关性分析等。本课程的目的是让每一位学生了解和掌握针对智能数据分析的一些基本思想、算法和工具，以便于为学生将来的研究和工作打下基础。

This course will provide an introduction to intelligent data analysis with special emphasis on fundamental data analysis tasks such as classification and clustering. Through lectures and labs, we will explore topics including classification and clustering algorithms, output calibration and model selection, data representation (feature analysis), learning-to-rank, correlation analysis, etc. The objective of this course is to enable each student to understand and master some basic ideas, algorithms and tools for intelligent data analysis tasks, so as to have a foundation for their further research and work.

9. 教学方法
Teaching Methods

理论课+实验课
32 lectures and 32 labs

10. 教学内容
Course Contents

Section 1	课程介绍 Introduction
Section 2	线性判别模型 Linear Discriminant Models
Section 3	支持向量机 Support Vector Machines
Section 4	神经网络 Neural Networks
Section 5	决策树 Decision Trees

	Section 6	生成模型 Generative Models
	Section 7	多分类器系统 Multiple Classifier Systems
	Section 8	输出矫正和模型选择 Output Calibration and Model Selection
	Section 9	机器学习理论 Theoretical Foundation of Machine Learning
	Section 10	聚类 Clustering
	Section 11	特征分析和降维 Feature Analysis and Dimensionality Reduction
	Section 12	排序学习 Learning to rank
	Section 13	图数据挖掘 Mining Graph Data
	Section 14	不均衡学习和代价敏感学习 Imbalanced Learning and Cost-sensitive Learning
	Section 15	半监督、主动和增量学习 Semi-supervised, Active and Incremental Learning
	Section 16	课程总结和回归 Summary and Review

11. 课程考核
Course Assessment

（①考核形式 Form of examination；②.分数构成 grading policy；③如面向本科生开放，请注明区分内容。If the course is open to undergraduates, please indicate the difference.）

平时作业 Assignments 30%
课程项目 Projects 20%
期末考试 Final Exam 50%

12. 教材及其它参考资料
Textbook and Supplementary Readings

T. Hastie, R. Tibshirani, J. Friedman,《The Elements of Statistical Learning: Data Mining, Inference, and Prediction》
Christopher M. Bishop,《Pattern Recognition and Machine Learning》